The empirical recurrence for OEIS A237163: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A237163 records "Empirical recurrence of order 59", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 549$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237163 is "Number of (n+1)X(4+1) 0..2 arrays with the maximum plus the minimum minus the upper median minus the lower median of every 2X2 subblock equal.". It has offset 1 and begins

$$2597, 20669, 165317, 1470317, 13373885, 126858645, 1234905853, 12370941453, \ldots$$

The entry states:

> Empirical recurrence of order 59 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 09:55:01, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 549$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 549$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1098$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{59} c_i\, a(n-i),$$

c_1	38	c_{16}	14117658629076	c_{31}	4159412279471724748	c_{46}	−444086661136863232
c_2	−371	c_{17}	−103158791332900	c_{32}	2078051147660241412	c_{47}	2319888053356117248
c_3	−2838	c_{18}	−129518825824655	c_{33}	−8164286576577101504	c_{48}	86410720616830464
c_4	64993	c_{19}	924072837087924	c_{34}	−3542304485928678000	c_{49}	−727960690761887232
c_5	−58567	c_{20}	964103900257693	c_{35}	13084030130740708600	c_{50}	−2463057123006720
c_6	−4394237	c_{21}	−6489848726510141	c_{36}	4824310506366988372	c_{51}	167819902964087040
c_7	15058024	c_{22}	−5838242601572712	c_{37}	−17077539997844469364	c_{52}	−3533160413067264
c_8	172006601	c_{23}	36216416422429550	c_{38}	−5201108564881985976	c_{53}	−27223090020771840
c_9	−838703867	c_{24}	28752450595327795	c_{39}	18071183306742752088	c_{54}	946858535276544
c_{10}	−4505621654	c_{25}	−162225277010758868	c_{40}	4378777606691183872	c_{55}	2907812839624704
c_{11}	26693317114	c_{26}	−115017716694513491	c_{41}	−15397149068880798752	c_{56}	−106061071417344
c_{12}	85503737174	c_{27}	587691474722981835	c_{42}	−2819908771789766592	c_{57}	−181780813578240
c_{13}	−572478983494	c_{28}	373164028715470753	c_{43}	10462825501680041280	c_{58}	4550416662528
c_{14}	−1237303411668	c_{29}	−1731047885680265093	c_{44}	1343176554987231904	c_{59}	5003075911680
c_{15}	8868339925003	c_{30}	−980075261566163264	c_{45}	−5598704889450166176		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59+4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

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